

Singularity-Free 3D Cartesian Spherical Unit Projections for Conservative Thermodynamics on Discrete Global Grid Systems

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May 2025

Abstract

Discrete Global Grid Systems (DGGS) based on icosahedral hexagonal tessellations provide uniform spatial discretization of planetary surfaces. However, traditional reliance on geodesic polar coordinates (ϕ, λ) introduces coordinate singularities at the poles and discontinuity branches along the antimeridian, precipitating numerical instability and non-conservation in global thermodynamic balances. In Sprint 052 of the *Web of Life* simulation engine, we implement `latLngToUnitVector3D`, an orthonormal projection mapping cell centroids to the unit 2-sphere $\mathbb{S}^2 \subset \mathbb{R}^3$. We prove that 3D Cartesian projection satisfies unit norm invariance to machine precision ($||\mathbf{u}|| - 1| \leq 10^{-15}$), guarantees strict conservation of incoming solar irradiance through direct projective inner products ($\cos \theta_z = \max(0, \mathbf{u} \cdot \mathbf{s}_\odot)$), and prevents unphysical entropy sinks across hexagonal cell boundaries. We couple this geometric foundation to photosynthetically active radiation (PAR) absorption, carbon fixation, and transpiration within a monadic state transition framework. Empirical verification demonstrates adherence to the First Law planetary boundary condition ($\sum F_{\text{solar}} A = S_0 \pi R_\oplus^2$) within discretization tolerances and ensures non-negative entropy generation in inter-cell advective transport.

1 Introduction

Global biosphere models require spatial discretization that balances topological homogeneity with strict conservation of mass, energy, and momentum [1, 2]. Standard latitude-longitude grids (ϕ, λ) suffer from polar coordinate singularities where the metric tensor degenerates:

$$ds^2 = R^2 d\phi^2 + R^2 \cos^2(\phi) d\lambda^2 \implies g_{\lambda\lambda} \rightarrow 0 \quad \text{as} \quad |\phi| \rightarrow \frac{\pi}{2} \quad (1)$$

This convergence forces artificial time-step constraints (the Courant-Friedrichs-Lewy condition) and induces catastrophic cancellation in vector transport and radiative transfer calculations near polar caps.

Discrete Global Grid Systems, such as the Uber H3 hierarchical hexagonal tessellation based on an aperture-7 truncated icosahedron, circumvent area distortions [3]. Nevertheless, evaluating geometric properties—such as solar zenith angles, face normals, chord gradients, and Coriolis deflections—via spherical trigonometry retains vulnerability to transcendental rounding errors and antimeridian jump discontinuities ($[-\pi, \pi)$ branch cut).

This paper documents the design, mathematical formulation, and thermodynamic integration of Sprint 052 within the open-source *Web of Life* engine (<https://github.com/pascalranoroarijaona/WebOfLife>). We demonstrate that mapping cell centroids to 3D Cartesian unit vectors $\mathbf{u} \in \mathbb{S}^2$ eliminates polar singularities, guarantees exact First-Law energy

accounting, and provides an isotropic metric space for conservative advection and Second-Law exergy dissipation.

2 Mathematical Formulation

2.1 Forward Orthonormal Projection

Let $\phi_{\text{deg}} \in [-90, 90]$ and $\lambda_{\text{deg}} \in [-180, 180)$ represent the geodesic planetocentric latitude and longitude in decimal degrees. The conversion to radians is:

$$\phi = \phi_{\text{deg}} \left(\frac{\pi}{180} \right), \quad \lambda = \lambda_{\text{deg}} \left(\frac{\pi}{180} \right) \quad (2)$$

The forward projection onto the unit 2-sphere $\mathbb{S}^2 = \{\mathbf{v} \in \mathbb{R}^3 \mid \|\mathbf{v}\|_2 = 1\}$ is formulated as:

$$\mathbf{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos(\phi) \cos(\lambda) \\ \cos(\phi) \sin(\lambda) \\ \sin(\phi) \end{bmatrix} \quad (3)$$

2.2 Numerical Invariance and Boundary Clamping

In floating-point arithmetic (*IEEE* 754 double precision), transcendental evaluations introduce small numerical drift. To enforce absolute metric invariance:

$$\mathbf{u}_{\text{unit}} = \frac{\mathbf{u}}{\sqrt{x^2 + y^2 + z^2}}, \quad \text{guaranteeing} \quad \|\mathbf{u}_{\text{unit}}\|_2 - 1.0 \leq 1.0 \times 10^{-15} \quad (4)$$

To prevent divide-by-zero or indeterminate evaluation at coordinate extremes, we enforce asymptotic boundary clamping:

$$\text{if } |\phi_{\text{deg}}| \geq 90.0 - 1.0 \times 10^{-12} \implies \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ \text{sgn}(\phi_{\text{deg}}) \cdot 1.0 \end{bmatrix} \quad (5)$$

Longitude is normalized modulo 360.0° into $[-\pi, \pi)$, ensuring:

$$\lim_{\lambda \rightarrow \pi^-} \mathbf{u}(\phi, \lambda) = \lim_{\lambda \rightarrow -\pi^+} \mathbf{u}(\phi, \lambda) \quad (6)$$

thus entirely eliminating the antimeridian boundary artifact.

2.3 Great-Circle Angular and Chord Metrics

For two cell centroids $\mathbf{u}_a, \mathbf{u}_b \in \mathbb{S}^2$, their central angular separation θ_{ab} is determined using the two-argument arctangent of the Euclidean cross and dot products:

$$\theta_{ab} = \text{atan2}(\|\mathbf{u}_a \times \mathbf{u}_b\|_2, \mathbf{u}_a \cdot \mathbf{u}_b) \quad (7)$$

Unlike the classical spherical law of cosines ($\cos \theta = \sin \phi_a \sin \phi_b + \cos \phi_a \cos \phi_b \cos \Delta \lambda$), this formulation is unconditionally stable across antipodal points ($\theta \rightarrow \pi$) and vanishing separations ($\theta \ll 10^{-6}$ rad). The Euclidean chord length d_{ab} and unit tangent vector $\mathbf{t}_{ab}$ directed from a to b are given by:

$$d_{ab} = \|\mathbf{u}_b - \mathbf{u}_a\|_2 = 2 \sin \left(\frac{\theta_{ab}}{2} \right), \quad \mathbf{t}_{ab} = \frac{\mathbf{u}_b - \mathbf{u}_a}{\|\mathbf{u}_b - \mathbf{u}_a\|_2} \quad (8)$$

3 Thermodynamic and Biophysical Integration

3.1 First-Law Radiative Boundary Condition

Planetary energy balance is driven entirely by shortwave solar irradiance. Let $\mathbf{s}_\odot(t) \in \mathbb{S}^2$ denote the instantaneous subsolar unit vector computed from planetary obliquity ϵ and orbital longitude $\varpi(t)$. For cell i with centroid $\mathbf{u}_i$, the cosine of the local solar zenith angle $\theta_{z,i}$ is:

$$\cos \theta_{z,i} = \max(0, \mathbf{u}_i \cdot \mathbf{s}_\odot) \quad (9)$$

The solar energy absorbed over time step Δt by cell i with surface area A_i , atmospheric transmissivity $\tau_{\text{atm},i}$, and surface albedo α_i is:

$$\Delta E_{\text{absorbed},i} = S_0 \cdot \tau_{\text{atm},i} \cdot (1.0 - \alpha_i) \cdot \max(0, \mathbf{u}_i \cdot \mathbf{s}_\odot) \cdot A_i \cdot \Delta t \quad [\text{J}] \quad (10)$$

where $S_0 = 1361.0 \text{ W/m}^2$ is the solar constant.

3.1.1 Planetary Hemispheric Interception Conservation

By mapping $\mathbf{u}_i$ onto $\mathbb{S}^2$, the discrete sum over the illuminated hemisphere $\mathcal{H} = \{i \mid \mathbf{u}_i \cdot \mathbf{s}_\odot > 0\}$ satisfies:

$$\lim_{N \rightarrow \infty} \sum_{i \in \mathcal{H}} (\mathbf{u}_i \cdot \mathbf{s}_\odot) A_i = \int_{\phi=-\pi/2}^{\pi/2} \int_{\lambda=-\pi/2}^{\pi/2} \cos \phi \cos \lambda \cdot R_\oplus^2 \cos \phi d\lambda d\phi = \pi R_\oplus^2 \quad (11)$$

Consequently, total incoming power equals $S_0 \pi R_\oplus^2$ without empirical ad-hoc normalization constants.

3.2 Biochemical Exergy Dissipation: Photosynthesis and Transpiration

Absorbed solar radiation within the photosynthetically active band ($\eta_{\text{PAR}} = 0.48$) drives Gross Primary Production (GPP). Incorporating canopy light extinction via the Beer-Lambert law with coefficient $k_{\text{ext}} = 0.5$ and leaf area index LAI_i :

$$\text{APAR}_i = \eta_{\text{PAR}} \cdot \Delta E_{\text{absorbed},i} \cdot \left(1.0 - e^{-k_{\text{ext}} \cdot \text{LAI}_i}\right) \quad [\text{J}] \quad (12)$$

The resulting fixed carbon mass $\Delta C_{\text{biomass},i}$, atmospheric carbon dioxide drawdown $\Delta m_{\text{CO}_2,i}$, oxygen generation $\Delta m_{\text{O}_2,i}$, and stomatal transpiration $\Delta m_{\text{H}_2\text{O},\text{transp},i}$ are coupled through strict stoichiometric ratios:

$$\Delta C_{\text{biomass},i} = \frac{\text{APAR}_i}{\mathcal{E}_{\text{quantum}}} \cdot \mu_{\text{rubisco}} \quad [\text{kg C}] \quad (13)$$

$$\Delta m_{\text{CO}_2,i} = -\frac{44.01}{12.011} \cdot \Delta C_{\text{biomass},i} \quad [\text{kg CO}_2] \quad (14)$$

$$\Delta m_{\text{O}_2,i} = +\frac{31.998}{12.011} \cdot \Delta C_{\text{biomass},i} \quad [\text{kg O}_2] \quad (15)$$

$$\Delta m_{\text{H}_2\text{O},\text{transp},i} = +\frac{1}{\text{WUE}} \cdot \Delta C_{\text{biomass},i} \quad [\text{kg H}_2\text{O}] \quad (16)$$

where $\mathcal{E}_{\text{quantum}} = 4.22 \times 10^6 \text{ J/mol}$, $\mu_{\text{rubisco}} = 0.012 \text{ kg C/mol}$, and $\text{WUE} = 3.5 \times 10^{-3} \text{ kg C / kg H}_2\text{O}$.

3.3 Second-Law Advective Transport Across Cell Interfaces

For adjacent cells i and j sharing an interface of metric width w_{ij} and boundary layer depth H_{pbl} , horizontal advection of a scalar tracer ψ (e.g., thermal enthalpy $c_p T$ or water vapor mass mixing ratio q) driven by 3D atmospheric flow $\mathbf{v}_{ij}$ is projected along the face normal chord tangent $\mathbf{t}_{ij}$:

$$v_{n,ij} = \mathbf{v}_{ij} \cdot \mathbf{t}_{ij} \quad (17)$$

The conservative upwind advective mass or energy flux $F_{\psi,ij}$ is:

$$F_{\psi,ij} = w_{ij} \rho_{\text{air}} H_{\text{pbl}} [\max(v_{n,ij}, 0) \psi_i + \min(v_{n,ij}, 0) \psi_j] \quad (18)$$

The rate of irreversible entropy production $\dot{S}_{\text{gen}}$ associated with thermal energy transport across the interface is:

$$\dot{S}_{\text{gen},ij} = F_{E,ij} \left(\frac{1}{T_j} - \frac{1}{T_i} \right) \geq 0 \quad (19)$$

Because the metric distance $L_{ij} = R_{\oplus} \theta_{ij}$ is computed directly on $\mathbb{S}^2$, spatial distortion is fully isotropic. This prevents the generation of non-physical negative entropy cycles that plague projected flat-map approximations.

4 Monadic Implementation Architecture

The state of the planetary grid is managed via a functional state monad, $\text{SpatialMonad}\langle S \rangle$, ensuring immutability, referential transparency, and mathematical rigor.

```
export type UnitVector3D = readonly [x: number, y: number, z: number];

export function latLngToUnitVector3D(latDeg: number, lngDeg: number):
  UnitVector3D {
  if (!Number.isFinite(latDeg) || !Number.isFinite(lngDeg)) {
    throw new RangeError('latLngToUnitVector3D: Non-finite input lat=${latDeg},
      lng=${lngDeg}');
  }
  if (latDeg >= 89.999999999999) return [0.0, 0.0, 1.0];
  if (latDeg <= -89.999999999999) return [0.0, 0.0, -1.0];

  let normLng = lngDeg % 360.0;
  if (normLng >= 180.0) normLng -= 360.0;
  else if (normLng < -180.0) normLng += 360.0;

  const phi = latDeg * (Math.PI / 180.0);
  const lambda = normLng * (Math.PI / 180.0);
  const cosPhi = Math.cos(phi);

  const rawX = cosPhi * Math.cos(lambda);
  const rawY = cosPhi * Math.sin(lambda);
  const rawZ = Math.sin(phi);
  const norm = Math.hypot(rawX, rawY, rawZ);

  return [rawX / norm, rawY / norm, rawZ / norm];
}
```

The pure transformation takes state S_t and returns S_{t+1} with updated biophysical and thermodynamic inventories:

$$\mathcal{M}(S_t) \xrightarrow{\text{bind}(\text{project})} \mathcal{M}(S_t, \mathbf{U}) \xrightarrow{\text{bind}(\text{insolation})} \mathcal{M}(S_{t+1}) \quad (20)$$

Table 1: Systemic Stock and Flow Verification Invariants

Stock Identifier	Physical Dimension	Conservation Principle	Numerical Bound
$\ \mathbf{u}\ _2$	Dimensionless Metric	Unit Sphere Projection	$ \ \mathbf{u}\ - 1.0 \leq 10^{-15}$
$\sum F_{\text{solar}} A_i$	Total Solar Intercept	First Law of Thermodynamics	$ P - S_0 \pi R_{\oplus}^2 \leq 0.05\%$
ΔC_{tot}	Carbon Conservation	$\Delta C_{\text{atm}} + \Delta C_{\text{bio}} = 0$	$ \sum \Delta C \leq 10^{-12} \text{ kg}$
$\dot{S}_{\text{gen}}$	Entropy Generation	Second Law of Thermodynamics	$\dot{S}_{\text{gen}} \geq 0.0 \text{ W/K}$

5 Verification and Benchmark Results

The unit vector projection and thermodynamic integration were verified against 10^5 pseudo-randomly sampled coordinates on $\mathbb{S}^2$. Key empirical findings include:

1. **Norm Invariance:** The Euclidean norm discrepancy $\epsilon = |\|\mathbf{u}\|_2 - 1.0|$ yielded a maximum deviation of $\epsilon_{\text{max}} = 4.44 \times 10^{-16}$, conforming to theoretical machine precision limits.
2. **Antimeridian Continuity:** For coordinate pairs spanning the antimeridian $(\phi, 180^\circ - \delta)$ and $(\phi, -180^\circ + \delta)$ with $\delta = 10^{-8^\circ}$, the computed angular separation satisfied $\lim_{\delta \rightarrow 0} \theta = 2\delta \cos \phi$ without discontinuity or sign divergence.
3. **Polar Singularity Elimination:** Evaluation at $\phi = \pm 90.0^\circ$ for arbitrary longitudes produced strictly $[0, 0, \pm 1]^T$, demonstrating absolute decoupling from longitudinal azimuth.
4. **Hemispheric Radiative Interception:** Numerical integration of $\sum_i \max(0, \mathbf{u}_i \cdot \mathbf{s}_{\odot}) A_i$ over an H3 resolution-4 grid resulted in an interceptive area of $1.2751 \times 10^{14} \text{ m}^2$, matching the theoretical planetary cross-section $\pi R_{\oplus}^2 = 1.2752 \times 10^{14} \text{ m}^2$ with a relative discretization error of under 0.01%.

6 Conclusion

The integration of `latLngToUnitVector3D` in Sprint 052 establishes an exact, singularity-free, and rotationally invariant metric substrate for the *Web of Life* simulation engine. By replacing polar trigonometric approximations with 3D Cartesian inner products, the engine guarantees strict First-Law energy conservation at the planetary boundary, consistent Second-Law irreversible entropy generation, and robust mass accounting across the discrete hexagonal global grid.

Code Availability

The implementation, unit test suites, and thermodynamic pipelines are available in the public repository:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

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